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A Controlled Smoking Cessation Trial for Substance-Dependent Inpatients

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Smoking treatment for newly recovering drug and alcohol-dependent smokers in a residential rehabilitation program was examined. The randomly assigned conditions (n = 50 each) were multicomponent smoking treatment (MST), MST plus generalization training of smoking cessation to drug and alcohol cessation (MST+G), or usual care (UC). Fifty participants who declined smoking treatment (treatment refusers) also were studied. Both treatment conditions achieved continuous smoking abstinence rates (MST: 12%, MST+G: 10%, at 12-month follow-up) that were significantly higher than in the UC condition (0%). The MST condition had a continuous drug and alcohol abstinence rate that was significantly higher than that of the MST+G condition (40% vs. 20% at 12-month follow-up) although neither differed significantly from that of the UC condition (33%). These results support the feasibility of smoking treatment for this population and provide information regarding appropriate treatment components.

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It has become increasingly clear that the negative health consequences of smoking among drug and alcohol abusers are substantial and the development of effective smoking treatments for this population has important positive health ramifications. Research has substantiated that an especially high percentage of drug and alcohol abusers smoke cigarettes (see Burling & Ziff, 1988; Hughes, 1995; Joseph, Nichol, Willenbring, Korn, & Lysaght, 1990), which places them at singular and combined risk for tobacco, alcohol, and other non-nicotine drug related diseases (Bien & Burge, 1990; Hurt et al., 1996; U.S. Department of Health and Human Services, 1989). Indeed, Hurt et al. (1996) recently found that tobacco smoking related diseases were the leading causes of premature morbidity and mortality among drug and alcohol abusers 20 years following substance abuse treatment.

One potential treatment option for this population is to provide concurrent smoking and drug and alcohol treatment in residential substance abuse treatment programs. Such settings provide ready access to large numbers of newly recovering substance abusers who may be motivated to improve their overall health. Furthermore, smoking treatment potentially could enhance the substance abuse treatment process by encouraging a positive overall change in lifestyle, promoting an increased sense of mastery or self-efficacy, and providing an opportunity for in vivo learning and practicing of relapse prevention and maintenance skills that could generalize to other abused substances (see Burling, Marshall, & Seidner, 1991; Hughes, 1995).

To date, surprisingly little empirical outcome research has been conducted to test the efficacy of concurrent smoking and drug and alcohol treatment. This has been partly due to the previously held beliefs that drug and alcohol abusers were not interested in quitting smoking, that the stress of quitting smoking too early in recovery would jeopardize their sobriety, or both (Bobo & Gilchrist, 1983; Sees & Clark, 1993). However, recent research suggests that even drug and alcohol abusers with as little as 1 month of abstinence are interested in (Irving, Seidner, Burling, Thomas, & Brenner, 1994) and capable of quitting smoking for brief periods of time without jeopardizing recovery from substance abuse (Bobo, McIlvain, Lando, Walker, & Leed Kelly, 1998; Burling et al., 1991; Hurt et al., 1994). In fact, a preliminary study conducted by Burling et al. (1991) found that smoking treatment had a small, positive effect on inpatient substance abuse treatment process and outcome (i.e., treatment retention, posttreatment substance use, and readmission rates) when delivered during the second month of inpatient treatment.

Unfortunately, the initial studies in this area have found the reductions in smoking to be relatively small and short term (e.g., see Bobo et al., 1998; Burling et al., 1991; Campbell, Wander, Stark, & Holbert, 1995; Hurt et al., 1994; Joseph, Nichol, & Anderson, 1993; Joseph et al., 1990; Saxton, McGuffin, & Walker, 1997). However, these studies have used either minimal intervention strategies or standardized treatments that may not adequately address the unique characteristics of a drug and alcohol population.

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In fact, there is a growing body of research that suggests that newly recovering drug and alcohol abusers are highly nicotine dependent smokers who may require additional or modified smoking interventions to address their particular biopsychosocial characteristics (see review by Burling, Ramsey, Seidner, & Kondo, 1997). Consequently, there has yet to be a satisfactory test of the efficacy and usefulness of smoking treatment for drug and alcohol abusers.

In this study we attempted to test the efficacy of two smoking treatments specifically designed for newly recovering drug and alcohol abusers in a residential rehabilitation program. One treatment (multicomponent smoking treatment [MST]) used multiple, intensive stop-smoking procedures to address the high level of nicotine dependence found in this population. The second treatment (multicomponent smoking treatment plus generalization training [MST+G]) used similar MST procedures and used the smoking cessation experience as an opportunity for "generalization training" of cessation and relapse prevention skills to drug and alcohol use. Posttreatment smoking and drug and alcohol outcomes were measured throughout the residential program and following discharge. These were compared with the outcomes of a usual care (UC) control group composed of inpatients who requested but did not receive the experimental smoking treatments and a treatment refuser (TR) comparison group composed of smokers who refused smoking treatment.

Method

Participants

The volunteer participants were 200 drug- and alcohol-dependent cigarette smokers in a residential rehabilitation program for homeless veterans at the Palo Alto Department of Veterans Affairs Health Care System. They were recruited from 417 consecutive admissions who smoked cigarettes and had a current *Diagnostic and Statistical Manual of Mental Disorders* (third ed., rev. [*DSM–III–R*]; American Psychiatric Association, 1987) diagnosis of a substance dependence disorder. To participate in the smoking study participants had to (a) complete the first 30 days of the inpatient program and not be in danger of imminent discharge (146 patients were excluded), (b) have smoked at least seven cigarettes per day for the previous 6 months and not regularly use any other form of tobacco (44 patients were excluded), and (c) have no medical conditions in which the use of nicotine patches was contraindicated (1 patient was excluded). Of the 226 substance-dependent smokers who met the study's eligibility criteria, 11 refused to participate, and 15 were excluded because the study conditions were temporarily closed or filled in accordance with the randomization procedure.

The final cohort consisted of 150 participants who volunteered for the study's smoking treatment (treatment accepters) and 50 who refused smoking treatment but agreed to participate in the study's assessments (TRs). The data in Table 1 describe the characteristics of the treatment accepters and refusers. The treatment accepter sample was predominantly male, middle-aged, racially diverse, socially disconnected, and relatively well educated. They had a modest number of various psychiatric problems. The treatment accepters tended to be long-term, highly nicotine-dependent cigarette smokers who were long-term abusers of alcohol, cocaine, or both. These characteristics were statistically similar in the TR sample, except that TRs reported significantly fewer prior quit-smoking attempts than treatment accepters (Kruskal–Wallis = 6.57, p = .01).

Procedures

Standard substance abuse treatment. All participants took part in an intensive, long-term, abstinence-oriented, 50-bed Veterans Affairs Domiciliary program that combined cognitive–behavioral (CB) skills training and therapeutic community (TC) procedures (see Burling, Seidner, Salvio, & Marshall, 1994). CB skills (e.g., assertion, conflict resolution, cognitive restructuring, "active listening," relapse prevention) were taught in classes

Table 1
Baseline Demographic and Background Characteristics for Sample

Variable	Treatment accepters (n = 150)	Treatment refusers (n = 50)
Demographics		
Male (%)	95	96
Age (*M, SD*)	39.61, 6.24	41.82, 8.15
Ethnicity (%)		
Caucasian	44	44
African American	45	46
Hispanic	5	10
Other	5	0
Marital status (%)		
Married	5	4
Separated	21	16
Divorced or widowed	43	50
Never married	31	30
Education (%)		
Less than high school/GED	12	10
High school/GED	33	38
More than high school/GED	55	52
Psychiatric problems		
No. Axis I and II SCID diagnoses (*Mdn*)	1	1
Cigarette smoking		
Cigarettes per day[a] (*M, SD*)	17.71, 7.39	16.54, 7.21
Nicotine level[a] (*M, SD*)	1.13, 0.22	1.12, 0.19
FTQ[b,c] (*M, SD*)	6.27, 1.81	5.73, 2.09
Carbon monoxide[b] (*M, SD*)	22.63, 8.22	21.28, 8.15
Years smoking[a] (*M, SD*)	23.68, 7.36	24.34, 9.20
Cigarette pack years[a] (*Mdn*)	24	22.5
No. prior quit attempts ≥24 hr[a] (*Mdn*)*	2	1
Drug and alcohol use		
Presenting drug problem[d] (%)		
Alcohol only	36	37
Cocaine only	29	39
Alcohol and cocaine	16	8
Other[e]	19	16
Years abusing this substance[d] (*M, SD*)	16.69, 8.24	16.69, 9.62
Prior inpatient treatments (*Mdn*)	2	2
No. prior quit attempts ≥24 hr (*Mdn*)	5	4

Note. GED = general equivalency diploma; SCID = Structured Clinical Interview for *DSM–III–R* (modified version; see *Data collection procedures and measures* section); FTQ = Fagerstrom Tolerance Questionnaire. [a] Treatment acceptors (TA; n = 149). [b] TA (n = 146). [c] Treatment refusers (TR; n = 48). [d] TR (n = 49). [e] For TAs, "Other" consisted of 6 participants who reported heroin only, 5 amphetamine only, 2 marijuana only, 1 other opiates only, 4 alcohol and heroin, 4 alcohol and cocaine, 4 alcohol and marijuana, 1 heroin and methadone, and 1 other opiates and marijuana. For TRs, 1 participant reported marijuana only, 2 alcohol and amphetamines, 2 alcohol and marijuana, 2 cocaine and amphetamines, and 1 heroin and alcohol.
* $p < .01$.

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and then practiced both inside the TC and in the community at large. The TC employed a patient self-government and a three-phase advancement system in which continued residence and additional privileges were contingent on demonstration of CB skill proficiency and progress toward treatment goals. Random and weekly breath alcohol tests and urine drug screens were conducted, and any drug or alcohol use resulted in discharge from the domiciliary. Program completion occurred when an individual had mastered the program-taught skills, obtained employment, saved enough funds for independent living, and secured housing. Successful program completion usually took 6–8 months, although the average length of stay for all patients admitted during the study period was 3.5 months.

Recruitment and randomization procedures. Potential volunteer participants were recruited for the smoking study by research staff after they completed the first 30 days of the residential program. Individuals who volunteered to participate and requested treatment were randomly assigned (n = 150; 50 per group) to one of the two treatment conditions (MST or MST+G) or the UC control condition. Individuals who did not want treatment but agreed to participate in the study's assessments were assigned to the TR condition.

MST and MST+G conditions. The two experimental smoking treatments were administered by master's- or doctoral-level counselors. Counselors initially received extensive group orientation training and subsequently participated in weekly group sessions involving supervisory training and treatment integrity monitoring. The weekly sessions were led by a doctoral-level supervisor and involved discussion of all active cases as well as feedback based on the supervisor's between-session review of at least one audiotape from each active case. Both conditions consisted of 5 weeks of prequit treatment and 4 weeks of postquit supportive counseling. The prequit treatment involved computerized nicotine fading (see Burling, Seidner, & Gaither, 1994) and daily, one-to-one counseling sessions (30–45 min/day) that included individualized contingency contracting[1] (Schwartz, 1987; Watson & Tharp, 1977) and relapse prevention training (Marlatt & Gordon, 1985; Stevens & Hollis, 1989). The first 2 weeks of postquit support involved daily counseling while participants used 14-mg nicotine patches, and the second 2 weeks involved biweekly counseling while participants used 7-mg patches. The patches were used to minimize participants' craving for cigarettes during their initial postquit period. The standard starting dose of 21 mg was not used because it was assumed that participants' nicotine intake had already been reduced by the computerized fading program (see Burling, Seidner, & Gaither, 1994).

The MST and MST+G groups were identical except that the MST program focused exclusively on smoking cessation, whereas the MST+G program used the smoking cessation experience as an opportunity for "generalization training" from cigarettes to drugs and alcohol. Thus, MST+G participants continually examined the similarities between successfully quitting smoking and drug and alcohol use, and they learned how each smoking cessation technique (e.g., contingency contracting) could be applied to quitting drugs and alcohol. During daily relapse prevention training, MST participants identified their high-risk situations for smoking and learned and practiced coping skills (both in session and *in vivo*) that could be used in these situations. In contrast, MST+G participants identified high-risk situations that were common to their smoking and drug and alcohol use, and they learned and practiced coping skills that could be used to prevent relapse to both cigarettes and drugs and alcohol.

UC and TR conditions. The UC and TR participants both received the standard inpatient program, but neither received any experimental smoking treatment. Both were told that they could quit smoking on their own or with the aid of any other smoking treatments, including those offered to all inpatients by means of centralized hospital-wide programs, prescription nicotine gum or patches available from the unit physician, or both. UC participants were informed of their right to receive an experimental smoking treatment after completing their participation in the study. Smoking was not otherwise addressed as a treatment issue for either group, although it was hospital policy that all inpatients restrict smoking to outside the program building.

Data collection procedures and measures. All data were collected by master's- or doctoral-level research staff. All follow-up interviews were conducted by research staff who did not serve as smoking counselors. All measures are available from Thomas A. Burling.

All potential participants completed a psychosocial interview and both the Structured Clinical Interview for *DSM–III–R*–Personality Disorders (SCID–II; Spitzer, Williams, Gibbon, & First, 1990a) and a modified SCID–P (patient edition) with Psychotic Screen (Spitzer, Williams, Gibbon, & First, 1990b) during the first week of the substance abuse treatment program. A structured psychosocial interview gathered preadmission information (lifetime, past year, and/or past 30 days) in the following areas: medical, living situation, employment, smoking, drugs and alcohol, legal, family–social, psychiatric, and overall quality of life. The instrument included all of the items composing the composite indices from the Addiction Severity Index (ASI; McLellan et al., 1988). The SCID–P (Axis I disorders with psychotic screen) was modified to assess only the 6 months prior to admission (rather than past month and lifetime presence) for the presence of major depressive and manic episodes and dysthymic and psychotic symptoms (rather than mood and psychotic disorders/syndromes) as well as 10 specific Axis I disorders. The complete SCID–II (personality disorders) was used (unmodified) to assess all 12 *DSM–III–R* personality disorders (including those not otherwise specified).

On entry into the smoking study, all participants completed a smoking history interview, provided breath and urine samples (to measure carbon monoxide [CO] and cotinine, respectively), and completed questionnaires about their attitudes toward smoking and drug and alcohol use (see below). The smoking history interview obtained data concerning current smoking behavior and smoking history. It included the Fagerstrom Tolerance Questionnaire (FTQ; Fagerstrom & Schneider, 1989) and items concerning cigarette pack years, quitting history, a rating of smoking's negative impact on one's health, the presence of 14 smoking-related physical symptoms, and attitude ratings (0–100) reflecting desire and intention to quit smoking and probability of quitting now and 6 months from now. A modified version of the Situational Confidence Questionnaire, short form (Annis, 1987), was used to assess confidence in ability to avoid using substances in 39 potentially high-risk situations. The modified version applied to alcohol and drugs (vs. alcohol only), rated confidence on a continuous scale of 0–100% (vs. 0%, 10%, 20% etc.), and rated ability to avoid using completely (vs. to avoid using heavily). An average confidence score across all situations was calculated, and the number of 0% ratings was computed. The Confidence Questionnaire short form (Baer & Lichtenstein, 1984; Condiotte & Lichtenstein, 1981), was used to assess confidence (0–100%) in ability to avoid smoking cigarettes in 14 potentially high-risk situations. An average score across all situations was calculated. The Reasons for Quitting Questionnaire (Curry, Wagner, & Grothaus, 1990) was used to assess motivations for quitting smoking. It consists of two 10-item subscales that assess intrinsic and extrinsic motives that are rated on a 5-point scale (0 = *not at all true* to 4 = *extremely true*). An intrinsic–extrinsic difference score was computed (Range: −4 to 4).

Beginning 1 week after participants made their first smoking cessation attempt (MST and MST+G conditions) or 6 weeks after they enrolled in the study (UC and TR conditions), and continuing weekly through the rest of their inpatient residency, participants completed a weekly assessment questionnaire that obtained 7-day point prevalence data concerning ciga-

[1] Participants developed individualized contracts that had smoking abstinence as the goal; were individualized; and contained specific, self-administered rewards for periods of abstinence and punishments for smoking. Participants also enlisted the assistance of a support person to help them implement their contract and problem solve difficulties when they quit smoking.

rette smoking (e.g., number of days smoked, number of cigarettes smoked per day). They also provided a breath sample for CO measurement.

At 1, 3, 6, and 12 months postdischarge from the inpatient program, all participants took part in paid follow-up assessments ($50 per interview) that included a structured interview and the provision of breath and urine samples (to measure CO and alcohol, and cotinine and drug use, respectively). The point prevalence follow-up interview obtained 30-day point prevalence data in the same areas as outlined for the psychosocial interview, and the items were comparable to those found in the psychosocial and smoking history interviews. The continuous-interval follow-up interview obtained data in the same areas as the point prevalence follow-up interview, but it contained fewer items and covered the entire time period between assessments (i.e., the time frame was "Since you were discharged . . . " or "Since your last follow-up . . . ," not "In the past 30 days . . . "). Interrater reliabilities for a sample of 4% of the completed point prevalence and continuous-interval follow-up interviews were extremely high: 99.31% and 98.01%, respectively.

Biochemical collection procedures. All biochemical assessments were obtained using standard, observed collection procedures. Expired air CO was assessed by having participants hold their breath for 20 s and then blow into a Vitalograph BreathCO (Vitalograph, Inc., Lenexa, Kansas) that provided a digital readout of exposure in parts per million. Breath alcohol was assessed by having participants blow into an Alco-Sensor IV (Intoximeters, Inc., St. Louis, Missouri) that provided a digital readout of percentage blood alcohol concentration. Urine samples for cotinine and drug use analyses were obtained by having participants urinate into a collection bottle. The urine samples were refrigerated and then transported to PharmChem Laboratories, Inc. (Menlo Park, California), according to standard protocols. Cotinine screening was conducted by radio immunoassay or enzyme multiplied immunoassay technique (EMIT) with confirmation by gas chromatography (GC). Cotinine–creatinine ratios were used in analyses to control for individual differences in urine volume. Drug use screening was conducted by EMIT with confirmation by GC (for amphetamines, barbiturates, benzodiazepine, methadone, cocaine metabolites, opiates, and phencyclidine) or high-performance thin layer chromatography (for cannabinoids).

Primary Outcome Analysis Procedures

Smoking outcomes were examined 1, 3, 6, and 12 months after participants (a) attempted to quit smoking, (b) withdrew from smoking treatment, (c) discontinued smoking treatment because of inpatient discharge, or (d) were 5 weeks post study enrollment (for UC and TR participants), as appropriate. This was accomplished by aggregating all of the smoking outcomes data collected during inpatient treatment and at the follow-ups conducted after residential treatment and then selecting the appropriate data points for each individual on the basis of the calendar dates of his or her assessments. Drug and alcohol outcomes were examined 1, 3, 6, and 12 months after participants were discharged from inpatient treatment. Analyses examined both smoking and drug and alcohol outcomes with respect to abstinence and relapse rates. In all cases, the MST and MST+G conditions were compared, both separately and combined, with the UC condition, to determine whether participants who received smoking treatment differed from those who did not. Then the MST and the MST+G conditions were compared to determine whether one treatment was superior to the other. Finally, the UC and TR conditions were compared to examine differences between participants who did and did not want smoking treatment among those who did not receive such treatment. Results are based on conservative analyses that included available data from all randomized participants (i.e., we used an "intent to treat" sample). As data were available for ≥90% of participants in virtually all analysis cells, and ≥80% in the remaining cells, no missing-data routines (e.g., treating missing participants as nonabstinent) were used.

Results

Baseline Comparisons

We determined whether the treatment accepter sample was representative of the potential participant pool, we assessed the effectiveness of the randomization procedure for the treatment accepter sample, and we determined whether there were differences between participants who did (UC) versus did not want smoking treatment (TR) among those who did not receive such treatment by conducting between-group comparisons on baseline participant characteristic variables (i.e., age, gender, ethnicity, marital status, education, number of Axis I and II psychiatric diagnoses, cigarettes per day, cigarette nicotine level, FTQ score, CO, years smoking, cigarette pack years, number of prior quit smoking attempts, presenting drug problem, years abusing this substance, number of prior inpatient treatments, and number of prior drug and alcohol quit attempts). The significance levels for all analyses were Bonferroni corrected within variable type (i.e., demographics, cigarette smoking, and drug and alcohol use). The treatment accepter sample was found to be representative of the potential participant pool, as the only observed difference between the 150 treatment accepters and the other 267 substance-dependent smokers admitted to the inpatient program during the study period was that the accepters were significantly younger, $t(415) = 2.86$, $p < .01$ (Ms = 39.61 vs. 41.57 years, respectively). (Note that these two groups could not be compared on FTQ score, cigarette pack years, CO level, or number of prior quit smoking attempts as these data were not available for nonparticipants.) The randomization procedure for the treatment accepter sample was found to be effective, as there were no significant differences among the participants in the various treatment accepter conditions (MST/MST+G combined vs. UC, MST vs. UC, MST+G vs. UC, and MST vs. MST+G). Finally, analyses found that there were no significant differences between the participants in the UC and TR conditions.

Treatment Process

Between-group comparisons were conducted on measures of smoking treatment process to determine whether there were any differences between participants in the two smoking treatment conditions (MST and MST+G). No significant between-group differences were found on variables related to the common components of treatment (see Table 2), such as the number and length of sessions, or participants' reasons for ending their smoking treatment. However, there was evidence that the critical expected difference between the two treatment groups was present. Two research associates (one bachelor's level and one master's level) who were blind to treatment condition assignment coded 81 audiotapes (40 MST and 41 MST+G) from randomly selected treatment sessions on a variety of variables, including the number of statements made concerning "generalization" (i.e., related to the similarity between quitting smoking and drugs and alcohol), the total number of statements made, and presumed treatment condition assignment. Analyses revealed that the ratio of generalization to total statements was significantly higher for the MST+G versus the MST group for both participant verbalizations, Kruskal–Wallis $\chi^2(1, N = 81) = 39.23$, $p < .0001$ (Mdn = .00 vs. .34), and counselor verbalizations, Kruskal–Wallis $\chi^2(1, N = 81) = 37.50$,

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Table 2
Smoking Treatment Process

Variable	MST (n = 50)	MST+G (n = 50)
Completed sessions (*M, SD*)	24.08, 11.79[a]	25.36, 10.85
Average length of sessions (min; *M, SD*)	33.53, 5.14[b]	34.60, 5.49[c]
Completed in vivo practices (*M, SD*)	13.60, 6.24[d]	14.32, 9.07[e]
Nicotine patches used (%)		
0–6	52	46
7–13	10	18
14–20	10	10
21–28	28	26
Reasons for ending smoking treatment (%)		
Completed entire 9-week smoking protocol	46	44
Withdrew from smoking treatment	18	10
Withdrew from smoking treatment and study	0	2
Prematurely discharged from inpatient program	36	44

Note. MST = multicomponent smoking treatment; G = generalization training.
[a] n = 49. [b] n = 36. [c] n = 39. [d] n = 45. [e] n = 47.

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$p < .0001$ (*Mdn* = .00 vs. .26). Also, 89% of the tapes were correctly classified in terms of their actual treatment condition assignment (83% of the MST and 93% of the MST+G tapes).[2]

Overall, nearly two thirds (62%) of all smoking treatment participants completed the 5-week prequit smoking treatment protocol and attempted to quit, and nearly half (45%) completed the entire 9-week protocol. Failure to complete the entire smoking treatment protocol was primarily due to premature termination from the inpatient drug and alcohol program (40%); only a small number of participants (15%) withdrew from the smoking treatment. Nearly three fifths (57%) of the participants who used at least one nicotine patch (n = 60) reported side effects (most commonly mild skin irritation/itching at the patch site, increased and/or vivid dreaming, mild headaches), but few discontinued (10%) or modified (2%) their patch use because of the side effects, and this was not a primary reason for discontinuing the smoking treatment.

Between-group comparisons also were conducted on length of inpatient stay and reasons for leaving inpatient treatment to determine whether the smoking treatments differentially influenced or interacted with inpatient treatment course (MST/MST+G combined vs. UC, MST vs. UC, MST+G vs. UC, MST vs. MST+G) and whether there were any differences related to wanting smoking treatment (UC vs. TR). No significant differences were found.[3]

Treatment Outcome

Smoking abstinence rates. Table 3 shows the CO-verified, 7-day point prevalence smoking abstinence rates by condition at 1, 3, 6, and 12 months postquit, and it shows the odds ratios for the five between-group comparisons that were conducted on these data. Overall 2 × 4 (Condition × Time) generalized estimating equation (GEE) analyses[4] found significant interactions for the three comparisons involving the treatment receiver and UC conditions: MST/MST+G combined versus UC, $\beta = -0.136$, $SE = 0.047$, $p < .01$; MST versus UC, $\beta = -0.179$, $SE = 0.061$, $p < .01$; MST+G versus UC, $\beta = -0.148$, $SE = 0.056$, $p < .01$. Follow-up logistic regression tests indicated that the treatment receivers had significantly higher abstinence rates than the UC participants only at the 1-month postquit point: MST/MST+G combined versus UC, $\chi^2(1, N = 148) = 23.74$, $p < .0001$; MST versus UC, $\chi^2(1, N = 99) = 25.25$, $p < .0001$; MST+G versus UC, $\chi^2(1, N = 98) = 13.92$, $p < .001$. Although enough UC participants achieved abstinence at the 3- to 12-month postquit assessments to result in no between-group differences at these points, inspection of the data indicated that UC participants achieved only short-term abstinence, whereas treatment receivers achieved sustained abstinence. For example, 2% versus 12% were abstinent at all four assessments, $\chi^2(1, N = 150) = 4.21$, $p < .05$. The overall 2 × 4 GEE analyses were not statistically significant for the MST versus MST+G or the UC versus TR comparisons.

As would be expected, among treatment receivers, analyses involving variables related to "treatment dose" (number of patches used, number of *in vivo* practices completed, etc.) and smoking outcome all revealed significant relationships. For example, logistic regression analyses indicated that abstinent participants had attended a greater number of smoking counseling sessions than nonabstinent participants had at all follow-ups: 1-month postquit,

[2] The tapes were coded as part of a substudy that has not yet been published. The coding system involved 11 content and 4 process dimensions as well as classification of treatment group assignment. Before coding the tapes used in analyses, the coders received approximately 20–25 hr of group training led by a doctoral candidate that involved discussion and refinement of the coding system and simultaneous but independent coding of tapes followed by resolution of discrepancies. Ten of the tapes used in analyses were independently coded by both raters, so interrater reliabilities could be determined (only the predesignated "primary" coder's ratings were used in the main analyses). These indicated that a relatively high level of agreement was achieved for most variables. For example, agreement was 90% for treatment group assignment.

[3] The overall median length of stay in the inpatient program was 112.50 days. The overall reasons for leaving the inpatient program were as follows: Thirty-nine percent successfully completed the program, 35% were asked to leave because of failure to comply with program rules, 22% left against medical advice, and 6.5% left for other reasons.

[4] GEE analyses (see Zeger & Liang, 1986; and Zeger, Liang, & Albert, 1988) are analogous to repeated measures analyses of variance but can be used when the dependent variable is categorical, as is the case here (i.e., abstinent vs. nonabstinent). The logit link and unspecified covariance options were selected for all GEE analyses.

Table 3
Postquit Smoking Abstinence Rates by Condition and Odds Ratios for Associated Between-Groups Comparisons

Condition	Time postquit (months) 1	3	6	12
	% Abstinent[a]			
Multicomponent smoking treatment (MST)	40[b]	18[c]	18[c]	19[d]
MST plus generalization training (MST+G)	27[c]	12[c]	11[d]	13[e]
Usual care (UC)	2[c]	11[d]	10[f]	13[g]
Treatment refusers (TR)	4[f]	13[g]	9[d]	9[d]
	Odds Ratios[i]			
Comparisons[h]				
MST/MST+G combined vs. UC[j]	24.00***	1.52	1.47	1.30
MST vs. UC[j]	32.00***	1.89	1.94	1.58
MST+G vs. UC[j]	17.30***	1.17	1.02	1.03
MST vs. MST+G	1.85	1.61	1.89	1.54
UC vs. TR	0.48	0.79	1.25	1.61

[a] Based on carbon monoxide verified, 7-day point prevalence abstinence rates at 1, 3, 6, and 12 months postquit (see *Primary Outcome Analysis Procedures* section). [b] $n = 50$. [c] $n = 49$. [d] $n = 47$. [e] $n = 45$. [f] $n = 48$. [g] $n = 46$. [h] Overall 2 × 4 (Condition × Time) comparisons were conducted using generalized estimating equation (GEE) analyses, and follow-up comparisons were conducted using logistic regression (LR) analyses. [i] Odds of the first condition listed being abstinent relative to the second. [j] $p < .01$ for GEE Condition × Time interaction.
*** $p < .001$ for follow-up LR.

$\chi^2(1, N = 98) = 47.65, p < .0001$; 3-months postquit, $\chi^2(1, N = 97) = 17.73, p < .0001$; 6-months postquit, $\chi^2(1, N = 96) = 12.95, p < .001$; 12-months postquit, $\chi^2(1, N = 91) = 10.40, p = .001$.

Smoking relapse rates. Figure 1 shows relapse rates based on CO-verified continuous abstinence at 1, 3, 6, and 12 months postquit. Survival analyses were significant for the three comparisons involving the treatment receiver and UC conditions: MST/MST+G combined versus UC, log rank and Wilcoxin $\chi^2(1, N = 150) = 19.07, p < .0001$; MST versus UC, log rank and Wilcoxin $\chi^2(1, N = 100) = 22.22, p < .0001$; MST+G versus UC, log rank and Wilcoxin $\chi^2(1, N = 100) = 13.79, p < .001$. Although the MST participants had consistently lower smoking relapse rates than the MST+G participants, the analyses comparing these two conditions were not significant. The analyses comparing the UC and TR conditions also were not significant.

Drug and alcohol abstinence rates. Table 4 shows breath alcohol- and urine drug test-verified 30-day point prevalence drug and alcohol abstinence rates by condition at 1, 3, 6, and 12 months postdischarge, and it shows the odds ratios for the five between-group comparisons that were conducted on these data. Overall 2 × 4 (Condition × Time) GEE analyses found significant main effects for time for all five comparisons, indicating that abstinence rates decreased over time for participants in all conditions: MST/MST+G combined versus UC, $\beta = 0.053$, $SE = 0.017$, $p < .01$; MST versus UC, $\beta = 0.052$, $SE = 0.021$, $p < .05$; MST+G versus UC, $\beta = 0.054$, $SE = 0.021$, $p < .01$; MST versus MST+G, $\beta = 0.054$, $SE = 0.021$, $p < .05$; UC versus TR, $\beta = 0.042$, $SE = 0.021$, $p = .05$. These analyses also revealed a significant main effect of condition for the MST versus MST+G comparison, $\beta = 0.781$, $SE = 0.327$, $p < .05$. Follow-up logistic regression tests indicated that the MST participants had significantly or near-significantly higher abstinence rates than the MST+G participants at all four assessments: $\chi^2(1, N = 96) = 3.36, p < .10$, at 1 month postdischarge; $\chi^2(1, N = 94) = 4.48, p < .05$, at 3 months; $\chi^2(1, N = 93) = 4.91, p < .05$, at 6 months; and $\chi^2(1, N = 85) = 4.27, p < .05$, at 12 months.

Drug and alcohol relapse rates. Figure 2 shows relapse rates based on breath alcohol- and urine drug test-verified continuous abstinence at 1, 3, 6, and 12 months postdischarge. Survival analyses were significant only for the comparison between the MST and MST+G conditions, log rank $\chi^2(1, N = 100) = 6.00$, $p < .05$, and Wilcoxin $\chi^2(1, N = 100) = 5.79, p < .05$, indicating that MST participants had significantly lower relapse rates than MST+G participants.

Other Outcome Analyses

Outcome analyses also were conducted on smoking and drug and alcohol use variables other than abstinence and relapse rates as well as an extensive list of variables from other content areas (i.e., medical–health, psychiatric, homelessness, employment, legal, family–social, quality of life, and confidence ratings). These data were collected 1, 3, 6, and 12 months after inpatient discharge, and they were analyzed by means of repeated measures analysis of variance when the data were continuous and normally distributed, using a significance level of $p < .05$. Otherwise, they were analyzed by means of appropriate between-group comparisons that used Bonferroni-corrected significance levels (i.e., $p < .0125$ as each variable was compared at each of the four time periods).

The only differences were that treatment receivers were employed for a greater percentage of the 30 days prior to their

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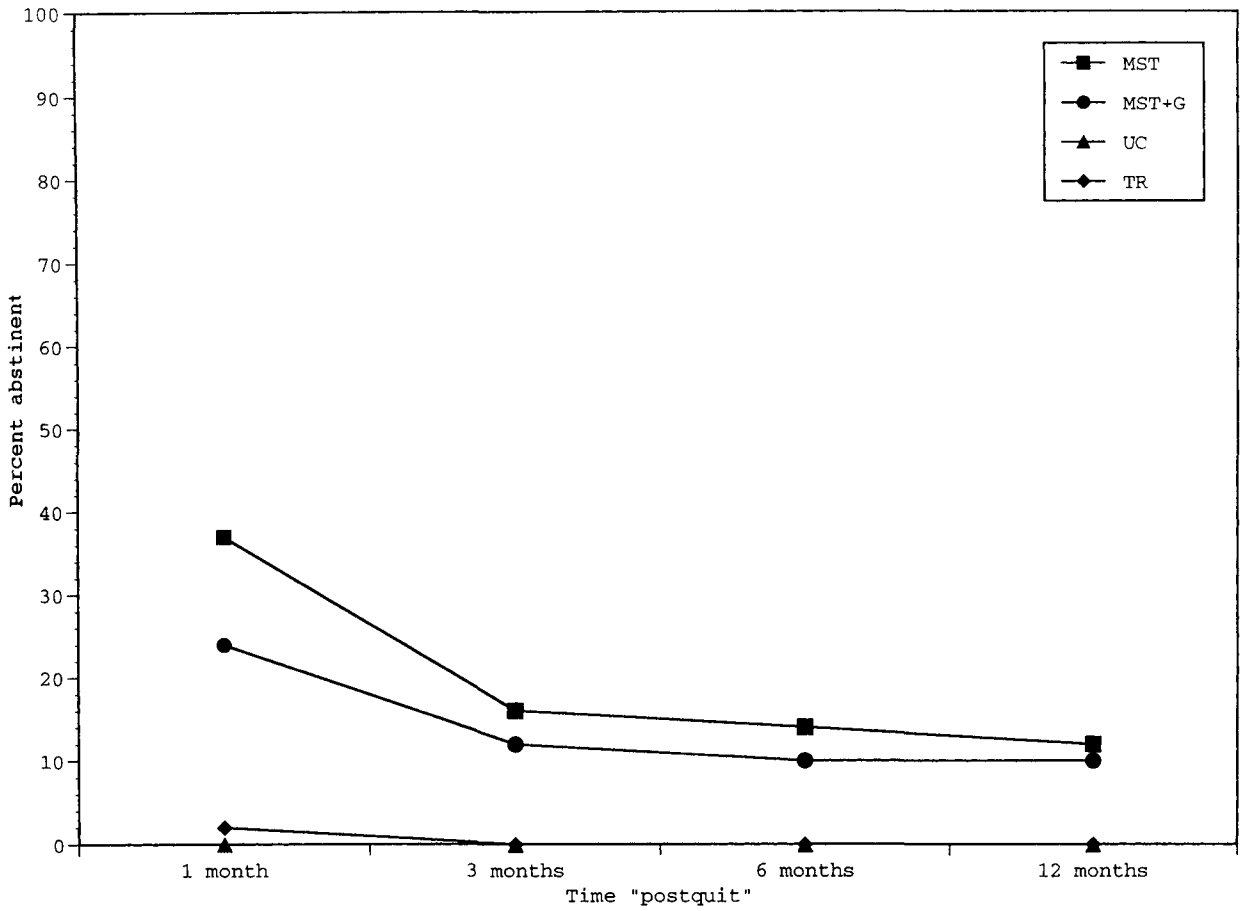

Figure 1. Survival curves for continuous smoking abstinence by condition. MST = multicomponent smoking treatment; G = generalization training; UC = usual care; TR = treatment refusers.

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3-month follow-up interview than the UC participants were: MST/MST+G combined versus UC, Kruskal–Wallis $\chi^2(1, N = 144) = 9.03, p < .01$; MST versus UC, Kruskal–Wallis $\chi^2(1, N = 95) = 6.74, p < .01$; MST+G versus UC, Kruskal–Wallis $\chi^2(1, N = 96) = 6.55, p = .01$ (*Mdn* = 100% vs. 30% for all three comparisons). Differences also were found indicating that MST participants had significantly lower (i.e., better) ASI Employment Index scores than MST+G participants at the 12-month assessment, Kruskal–Wallis $\chi^2(1, N = 90) = 7.83, p < .01$ (*Mdn* = .29 vs. .70).

Discussion

This is one of the most comprehensive, controlled clinical trials to assess the impact of smoking treatment designed for newly recovering drug- and alcohol-dependent smokers to date. The primary finding was that inpatient drug and alcohol dependent smokers were both willing and able to quit smoking during their inpatient residency without jeopardizing their substance abuse treatment or posttreatment drug and alcohol abstinence. This finding replicates previous work (e.g., see Burling et al., 1991; Campbell et al., 1995; Hurt et al., 1994; Joseph et al., 1990) and stands in direct contrast to the belief previously held by some that drug and alcohol abusers should not attempt to quit smoking during the initial stages of their treatment for substance abuse.

The smoking quit rates observed in the present study compare quite favorably to those of other smoking studies with inpatient substance abusers (Bobo et al., 1998; Burling et al., 1991; Campbell et al., 1995; Hurt et al., 1994; Joseph et al., 1993; Saxton et al., 1997) and represent a substantial improvement over the abstinence rates found in the preliminary study by Burling et al. (1991). This is especially encouraging given that the criteria for abstinence in the present study were very strict and our results were obtained with a very impaired population of homeless veterans who were highly nicotine dependent and had very recent and severe substance abuse histories. To this end, the results of the present study represent a very rigorous test of the ability of recovering substance abusers to quit smoking.

That the present study achieved improved smoking quit rates relative to the initial study (Burling et al., 1991) was likely due to the fact that the present study used longer counseling sessions (30 min/day vs. 15 min/day), provided specific relapse prevention training, continued counselor support longer after the initial quit attempt (4 weeks vs. 10 days), and used nicotine patches to reduce craving following smoking cessation. Similarly, the residential program in the present study restricted smoking to specific outside areas and times, had a

Table 4
Postquit Drug and Alcohol Abstinence Rates by Condition and Odds Ratios for Associated Between-Groups Comparisons

Condition	Time postdischarge (months) 1	3	6	12
	% Abstinent[a]			
Multicomponent smoking treatment (MST)	77[b]	67[c]	67[c]	61[d]
MST plus generalization training (MST+G)	59[e]	46[f]	45[b]	39[g]
Usual care (UC)	61[c]	63[c]	67[h]	51[i]
Treatment refusers (TR)	69[h]	63[d]	73[g]	57[j]
	Odds Ratios[l]			
Comparisons[k]				
MST/MST+G combined vs. UC	1.35	.76	.63	.93
MST vs. UC	2.10	1.21	1.03	1.49
MST+G vs. UC	.93	.50	.40*	.60
MST vs. MST+G[m]	2.26	2.44*	2.56*	2.48*
UC vs. TR	.70	.98	.75	.77

[a] Based on breath alcohol- and urine drug test-verified, 30-day point prevalence abstinence rates at 1, 3, 6, and 12 months postdischarge from inpatient treatment. [b] $n = 47$. [c] $n = 46$. [d] $n = 41$. [e] $n = 49$. [f] $n = 48$. [g] $n = 44$. [h] $n = 42$. [i] $n = 42$. [j] $n = 40$. [k] Overall 2 × 4 (Condition × Time) comparisons were conducted using generalized estimating equation (GEE) analyses, and follow-up comparisons were conducted using Logistic Regression (LR) analyses. Only significant findings involving condition are shown as these are the focus of this study. [l] Odds of the first condition listed being abstinent relative to the second. [m] $p < .05$ for GEE condition main effect.
* $p < .05$ for follow-up LR.

specific relapse prevention class for substance abuse that likely generalized to smoking, sponsored a variety of sports teams that likely encouraged positive health attitudes and behaviors (see Burling, Seidner, Hanser, & Marshall, 1992), had more peer support for smoking cessation, and had a smaller percentage of substance abuse counselors who smoked and thus tacitly condoned smoking. On the other hand, certain characteristics of our study (e.g., low power for analyses of smoking abstinence rates beyond the 1-month follow-up) and treatment setting (e.g., a high level of peer support, which may have fostered some contamination across conditions) may have decreased the likelihood of finding differences among the experimental conditions. Indeed, with respect to contamination, informal discussions with treatment participants indicated that some information was shared with UC participants, and that this likely resulted in the sporadic quitting noted among these individuals. On a related note, although one might speculate that exposure to additional smoking treatments during the follow-up period might account for the abstinence observed in the UC group, such exposure was minimal, and none of the UC participants who were classified as abstinent reported such exposure.

Although the overall smoking outcomes in the present study were markedly better than those of Burling et al.'s (1991) initial study, they were still lower than those typically found with nonabusing populations who receive intensive, multicomponent treatments (30%–50%; e.g., see Brown & Emmons, 1991; Schwartz, 1987). This could partially be due to the fact that our smokers were highly nicotine dependent and thus more likely to have a difficult time quitting smoking and avoiding relapse. This also could be due to the fact that there are a number of factors inherent to the residential treatment setting that could potentially interfere with smoking cessation efforts (see Burling, Ramsey, et al., 1997). For example, there was daily exposure to active smokers and smoking staff who could have passively or actively interfered with smoking cessation efforts. It appears to be clear that drug and alcohol abusers need especially intensive smoking cessation treatments and a treatment environment that is highly supportive of smoking cessation.

Smoking cessation rates are likely lowered by the fact that individuals who relapse to drugs and alcohol are unlikely to quit smoking. Thus, newly recovering substance abusers are in fact required to change two behaviors, whereas nonabusers are required to change only one. It is not surprising that the 1-year continuous smoking abstinence rates of participants in the present study who remained continuously abstinent from drugs and alcohol were substantially higher (MST: 29%, MST+G: 50%) than the rates of participants who relapsed to drugs and alcohol (MST: 0%, MST+G: 3%). Furthermore, the smoking cessation rates among these individuals actually approximate those typically found among nonabusers who receive intensive treatments (30%–50%; e.g., see Brown & Emmons, 1991; Schwartz, 1987), suggesting that drug and alcohol abusers may be as likely to quit smoking as nonabusers when the impact of substance abuse is factored out (cf. Martin et al., 1997; Patten, Martin, Myers, Calfas, & Williams, 1998). Practically, this suggests that future smoking cessation efforts with this population should be more clearly designed to minimize the potential interference of the sequelae of substance abuse.

Our finding that the MST+G group had poorer smoking and drug and alcohol outcomes than the MST group was unexpected and difficult to interpret given that neither smoking treatment condition differed from the UC condition. A possible explanation for this finding is that the content of the MST+G treatment (i.e.,

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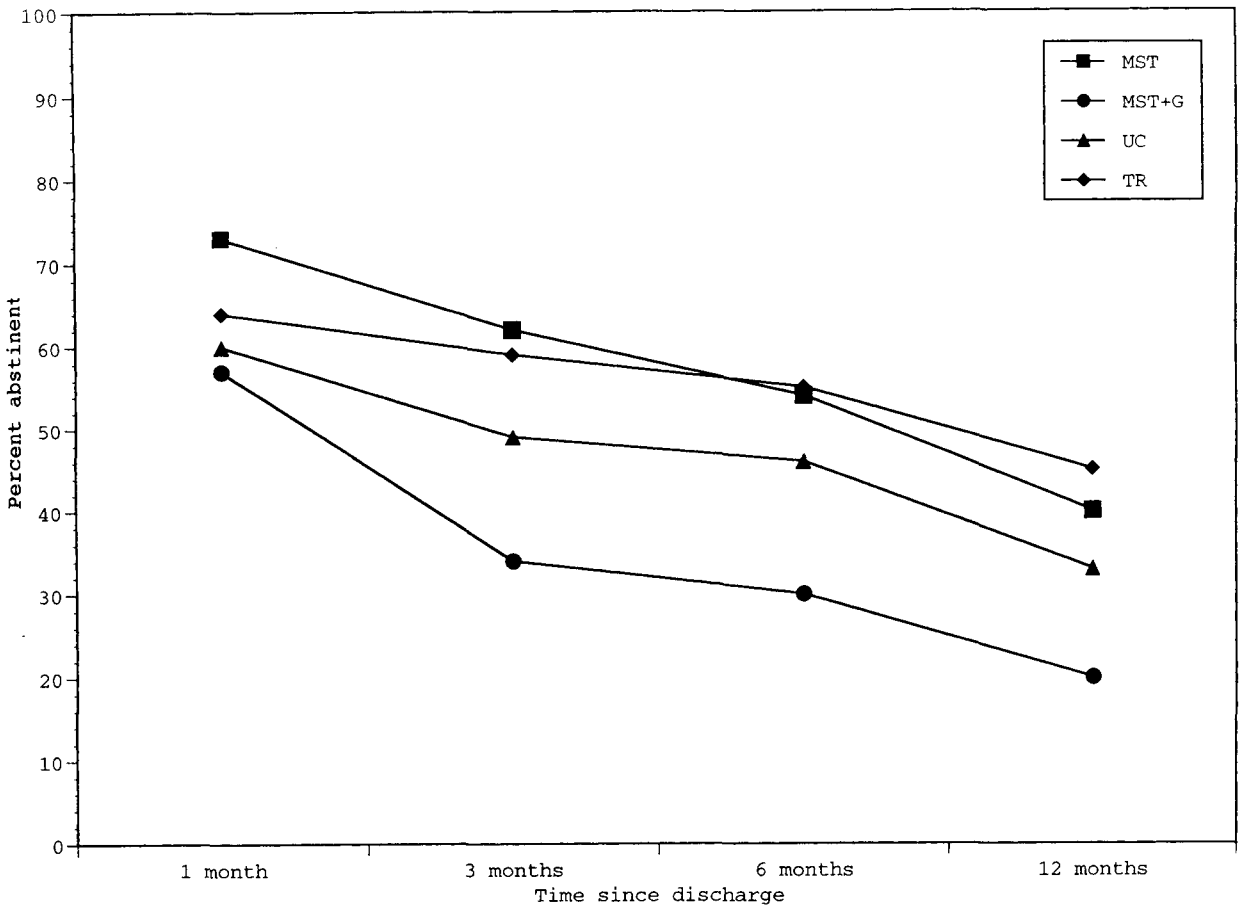

Figure 2. Survival curves for continuous drug and alcohol abstinence by condition. MST = multicomponent smoking treatment; G = generalization training; UC = usual care; TR = treatment refusers.

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generalization) was too obvious to interest or engage the participants or that discussions of generalization consumed too much treatment time relative to other important treatment issues. Support for the idea that the content of the MST+G group was too obvious comes from intake questionnaires in which participants reported that nicotine and drugs and alcohol were similar (see Irving et al., 1994) and from counselor observations that participants in the MST group frequently made spontaneous observations about the similarities between smoking and drug and alcohol use. Another possibility is that the MST+G condition placed too much emphasis on the similarities between smoking and drugs and alcohol such that participants became discouraged and/or experienced lowered self-efficacy to avoid drugs and alcohol if they did poorly in the smoking program. This may suggest that the content of the smoking program can potentially affect substance abuse treatment in negative ways and that care should be taken to not inadvertently place too much emphasis on using success of quitting smoking as a marker of success in drug and alcohol abuse.

It should be noted that the present study used a highly intensive and individualized treatment that may not be suitable for or practical in health care settings with more limited resources. However, the goal of the present study was to demonstrate the feasibility and safety of smoking treatment for drug and alcohol abusers. We hope that the positive results of the present study in this regard will promote the development of less costly and more effective interventions. It also is important to note that the specific nature of our population (homeless, substance dependent veterans) limits the generalizability of the study and requires replication with other drug and alcohol dependent populations. Finally, it was unfortunate that our sample was too small and heterogeneous to afford sufficient power to conduct more detailed comparisons of outcome between the unique substance-related disorders (e.g., alcohol abusers vs. cocaine abusers). Such analyses could provide important information, as there is increasing evidence that smokers with different substance-dependence disorders may differ on important dimensions (see Burling, Ramsey, et al., 1997; Burling, Salvio, Seidner, & Ramsey, 1997). Overall, the results of the present study are encouraging and informative. We safely achieved a modest smoking cessation rate among a highly nicotine dependent and impaired population of newly recovering, drug- and alcohol-dependent homeless veterans. Furthermore, we observed differences in outcome between our two treatment groups that appear to suggest that smoking and drug and alcohol treatment may best be delivered in a parallel, but not combined, format. These data provide an important foundation from which other smoking cessation treatments for drug and alcohol abusers can be designed and tested.

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